

From Classical ML to Quantum: A Novel Quantum Support Vector Classifier for Four-Class EEG Emotion Recognition on the DEAP Dataset

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Declaration

We, **Naina Rai(BT23CSD011), Tsewang Stenzin(BT23CSD017), Akshat Jain(BT23CSD020), Ritesh Tyagi(BT23CSD030)**, hereby declare that this project work titled **“From Classical ML to Quantum: A Novel Quantum Support Vector Classifier for Four-Class EEG Emotion Recognition on the DEAP Dataset”** is carried out by us in the Department of Computer Science and Engineering of Indian Institute of Information Technology, Nagpur. The work is original and has not been submitted earlier whole or in part for the award of any other certification programme at this or any other Institution /University.

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Date: 17 November 2025

Certificate

This is to certify that the project titled “**From Classical ML to Quantum: A Novel Quantum Support Vector Classifier for Four-Class EEG Emotion Recognition on the DEAP Dataset**”, submitted by **Naina Rai(BT23CSD011), Tsewang Stenzin(BT23CSD017), Akshat Jain(BT23CSD020), Ritesh Tyagi(BT23CSD030)** in partial fulfillment of the requirements for the Mini-Project in CSE department IIIT Nagpur. The work is comprehensive, complete and fit for final evaluation.

Date: 17 November 2025

Mr. Pravin Satyavan Bhagat
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Abstract

Accurate emotion recognition from electroencephalogram (EEG) signals remains a significant challenge in computational neuroscience and human-computer interaction, with profound implications for adaptive HCI and future mental health diagnostics. This paper presents a comprehensive comparative analysis of classical, deep learning, and quantum machine learning (QML) approaches to tackle this problem. We focus on classifying four distinct emotional states (Happy, Relaxed, Sad, and Anger) from the public DEAP dataset, providing a direct comparison of these three computational paradigms on a complex, multi-class task.

To make this high-dimensional data tractable, we first engineer a low-dimensional feature set from the raw EEG data. This set comprises six key features: Age, Alpha_F3, Beta_P4, Theta_Cz, Entropy_Pz, and FA_F3F4. Classical machine learning models, including K-Nearest Neighbors, XGBoost, and Logistic Regression, established a baseline accuracy in the 41-46% range, a result that underscores the inherent non-linear difficulties of the feature space.

A deep learning hybrid CNN-LSTM model, designed to capture spatio-temporal dynamics, achieved a significantly higher accuracy of 74.33%. This result is highly competitive, even outperforming contemporary models like 3D-EEGU-Net, which reported 72.02% on a simpler ternary classification task.

We then introduce a novel quantum-hybrid model by applying a Quantum Support Vector Classifier (QSVC). This method required a dimensionality reduction step where the 6 features were compressed to 4 principal components, making them suitable for encoding onto a quantum feature map. This QSVC model achieved a state-of-the-art accuracy of 98.83%, with near-perfect macro-averaged Precision, Recall, and F1-Scores of 0.99.

This striking finding dramatically outperforms both the classical and advanced deep learning baselines. It provides a strong proof-of-concept, demonstrating the significant potential of quantum machine learning kernels to find near-perfect decision boundaries when applied to complex, low-dimensional biosignal data.

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List of Symbols, Abbreviations

Abbreviation	Full Form
CNN	Convolutional Neural Network
DE	Differential Entropy
DEAP	A Database for Emotion Analysis using Physiological signals
EEG	Electroencephalogram
FA	Frontal Asymmetry
FQK	Fidelity Quantum Kernel
HCI	Human-Computer Interaction
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory
NISQ	Noisy Intermediate-Scale Quantum
PCA	Principal Component Analysis
PSD	Power Spectral Density
QKE	Quantum Kernel Estimation
QML	Quantum Machine Learning
QPU	Quantum Processing Unit
QSVC	Quantum Support Vector Classifier
SVM	Support Vector Machine

1. INTRODUCTION

Emotion is a complex psychological and physiological state that encompasses sensation, thought, and behavior, playing a critical role in daily life. The accurate recognition of human emotion is a significant goal in computational neuroscience and computer science, with profound implications for mental health diagnostics, cognitive psychology, and the development of intelligent and humane human-computer interaction (HCI) systems. Effective emotion detection is particularly critical for monitoring the mental well-being of individuals in isolated or high-stress environments, such as astronauts or submariners, where prolonged confinement can negatively impact mental health and work efficiency.

Traditionally, emotion recognition has relied on external characteristics such as speech, facial expressions, and body movements. While accessible, these methods are "susceptible to deliberate disguises, failing to accurately capture true emotional states". In contrast, physiological indicators, especially electroencephalogram (EEG) signals, offer a more objective and reliable alternative. EEG signals directly reflect the neural electrical activity associated with emotional processes and are difficult to consciously manipulate or disguise. With its excellent temporal resolution, EEG can capture the dynamic, real-time changes in brain activity, making it a powerful tool for affective analysis. To facilitate standardized research in this area, public datasets such as the **DEAP (A Database for Emotion Analysis using Physiological signals)** have been developed. The DEAP dataset, in particular, is a widely used multimodal benchmark, providing EEG and peripheral physiological signals from 32 participants as they watched 40 one-minute music videos to elicit various emotional states .

Despite its promise, analyzing EEG data for emotion recognition presents significant challenges. First, EEG signals are high-dimensional, inherently noisy, and vary significantly between individuals. Second, deep learning models, while powerful, often require large datasets to train effectively, which is difficult and costly to acquire for EEG signals. Much of the current research, therefore, focuses on two key areas: (1) developing robust feature extraction methods and (2) designing advanced classification models.

In recent years, deep learning models like Convolutional Neural Networks (CNNs) , Deep Belief Networks (DBNs) , and Long Short-Term Memory (LSTM) networks have been applied to this problem. For instance, a recent study by Huang et al. proposed a 3D-EEGU-Net model that treated EEG data as medical images, achieving a notable **72.02%** accuracy on a 3-class (ternary) valence classification task on the DEAP dataset. This result highlights the capability of modern deep learning architectures.

This paper presents a comparative analysis of classical, deep learning, and quantum machine learning approaches for a 4-class emotion classification task (Happy, Relaxed, Sad, and Anger) using the DEAP dataset. Our contributions are threefold:

1. We engineer a low-dimensional, 6-feature set from the raw 32-channel EEG data, incorporating bandpower (Alpha_F3, Beta_P4, Theta_Cz), signal entropy (Entropy_Pz), frontal asymmetry (FA_F3F4), and participant metadata (Age).
2. We establish baseline performance on this feature set using classical machine learning (Logistic Regression, KNN, XGBoost) and deep learning (a standard CNN and a hybrid CNN-LSTM) models.
3. We introduce a hybrid Quantum Support Vector Classifier (QSVC) model. This approach first applies Principal Component Analysis (PCA) to reduce our feature set to 4 dimensions, which are then encoded into a quantum feature map to train the classifier.

Our results demonstrate a clear hierarchy of model performance. The classical models served as a baseline with accuracies in the 41-46% range. Our deep learning CNN-LSTM model achieved a strong accuracy of **74.33%**, outperforming the 3D-EEGU-Net baseline on a more complex 4-class task. Most significantly, our proposed QSVC model achieved a state-of-the-art accuracy of **98.83%**, with macro-averaged Precision, Recall, and F1-Scores of 0.99. This finding dramatically outperforms all other approaches, highlighting the immense potential of quantum machine learning for classifying complex, low-dimensional biosignal data.

2. Literature Review

In recent years, the field of EEG-based emotion recognition has attracted considerable research, yielding significant results. Research efforts are generally bifurcated into two main domains: (1) the engineering of robust features from the complex, high-dimensional EEG signals, and (2) the development of sophisticated classification models to interpret these features.

Traditional machine learning approaches have served as a foundational baseline. Studies have explored a variety of handcrafted features, including Hjorth parameters, Power Spectral Density (PSD), Differential Entropy (DE), and various time-frequency features. These features are often fed into classical classifiers. For example, Li et al. used Hjorth features with a Support Vector Machine (SVM) to achieve **83.33%** accuracy on the SEED dataset. On the DEAP dataset, models such as K-Nearest Neighbors (KNN) and Random Forests (RF) have been used, achieving accuracies in the 62-63% range. While computationally efficient, these methods often fail to capture the complex, non-linear, and spatiotemporal relationships inherent in EEG data.

To address these limitations, deep learning models have become increasingly prevalent. Deep Belief Networks (DBNs) were proposed to capture reliable emotional stage transitions from EEG features. More commonly, Convolutional Neural Networks (CNNs) have been used to extract spatial features from mapped EEG channels. Tong et al. , for instance, used a CNN on the DEAP dataset to achieve **71.70%** accuracy. Hybrid models combining CNNs with Long Short-Term Memory (LSTM) networks have also been explored to capture both spatial and temporal dynamics. Hima et al. proposed a CNN-LSTM model using DE features that achieved **72.38%** accuracy on the DEAP dataset. In a similar vein, Huang et al. recently proposed a 3D-EEGU-Net, treating multichannel EEG features as a 3D image, which achieved **72.02%** on a 3-class valence task on the DEAP dataset. These results show a consistent performance ceiling for advanced deep learning models in the low 70% range on this task.

While deep learning has shown an advantage over classical methods, it typically requires substantial data and computational resources. A nascent but highly promising field, Quantum Machine Learning (QML), offers a new paradigm for data classification. QML

leverages the principles of quantum mechanics to perform computations in a high-dimensional Hilbert space. Specifically, Quantum Support Vector Classifiers (QSVC) use quantum kernels to map classical feature vectors into a quantum state. This "kernel trick" allows for the identification of complex patterns that may be inseparable by classical or standard deep learning models. While QML has been explored for other classification tasks, its application to the complex, low-dimensional feature space of EEG-based emotion recognition is not yet widely established.

This paper bridges this gap by first establishing robust baselines using both classical (Logistic Regression, KNN, XGBoost) and deep learning (CNN, CNN-LSTM) models on a low-dimensional, 6-feature set derived from the DEAP dataset. We then introduce a hybrid QSVC model, applying PCA for feature reduction before encoding the features using a Qiskit [ZZFeatureMap](#). This allows for a direct comparison of classical, deep learning, and quantum approaches on the same feature-engineered data, evaluating their efficacy for this complex biosignal classification problem.

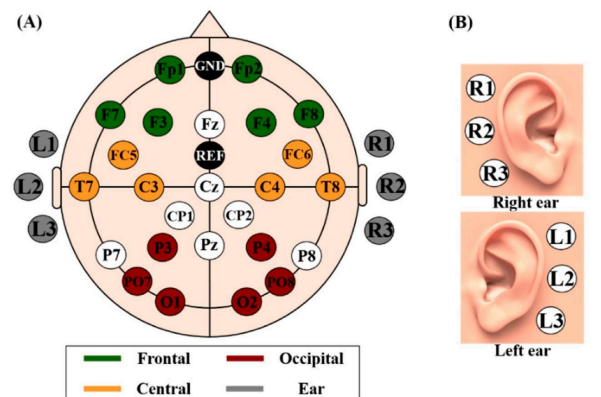
3. METHODOLOGY

This study was designed to compare the performance of classical, deep learning, and quantum machine learning models on an EEG-based emotion classification task. All models were trained and tested on the same foundational dataset, which was derived from the raw DEAP dataset through a standardized feature engineering pipeline.

3.1. DEAP Dataset and Preprocessing

To validate the proposed models, we employed the public **DEAP (A Database for Emotion Analysis using Physiological signals)** dataset. This dataset is a widely used multimodal benchmark for affective computing. It contains data from 32 participants (16 male, 16 female) who each watched 40 one-minute music videos intended to elicit different emotional states.

The raw data for each trial consists of 63 seconds of signal, including a 3-second pre-stimulus baseline. The data was recorded from 40 channels: 32 EEG channels (arranged according to the 10-20

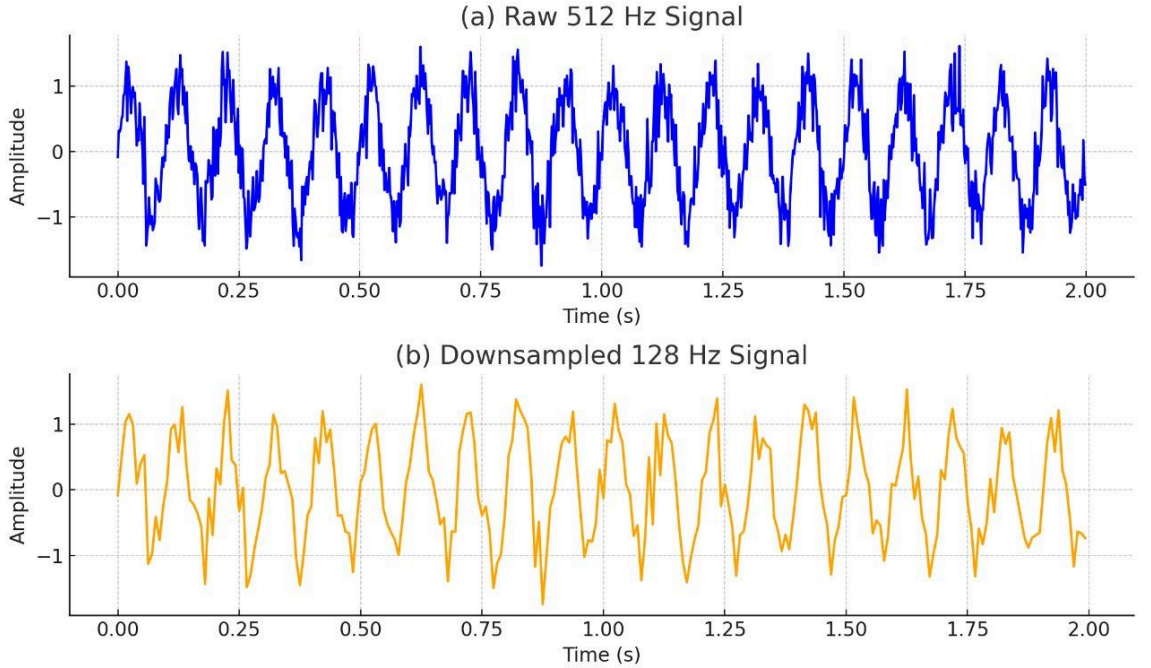


international system) and 8 peripheral physiological channels. For this study, we focused exclusively on the 32 EEG channels listed in TABLE I.

Array Name	Array Shape	Description
data	$32 \times 40 \times 32 \times 8064$	participant $\times$ video/trial $\times$ EEG $\times$ data
labels	$32 \times 40 \times 2$	participant $\times$ video/trial $\times$ (valence, arousal)

TABLE I: THE 32 EEG CHANNELS USED FROM THE DEAP DATASET

In the preprocessing stage, the raw 512 Hz EEG signals were first downsampled to **128 Hz**. This step reduces computational load while preserving the frequency bands most relevant to emotional analysis (typically 4-45 Hz).



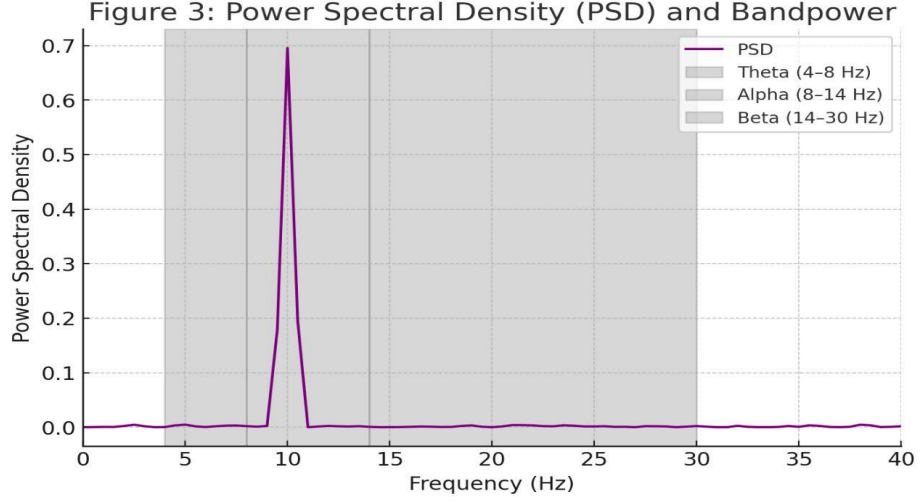
3.2.Feature Engineering

Instead of using the full time-series data or complex spectral images, we engineered a low-dimensional feature set to represent the brain's state during each trial. For each of the 40 trials for all 32 participants, we extracted the following features:

1. **Bandpower Features:** We calculated the power spectral density (PSD) using Welch's method and then computed the average power within specific frequency

bands. This is represented by equation (1), the integration of the PSD(f) over the frequency band from f_{low} to f_{high} :

$$P_{band} = \int_{f_{low}}^{f_{high}} PSD(f) df, \quad (1)$$



We extracted:

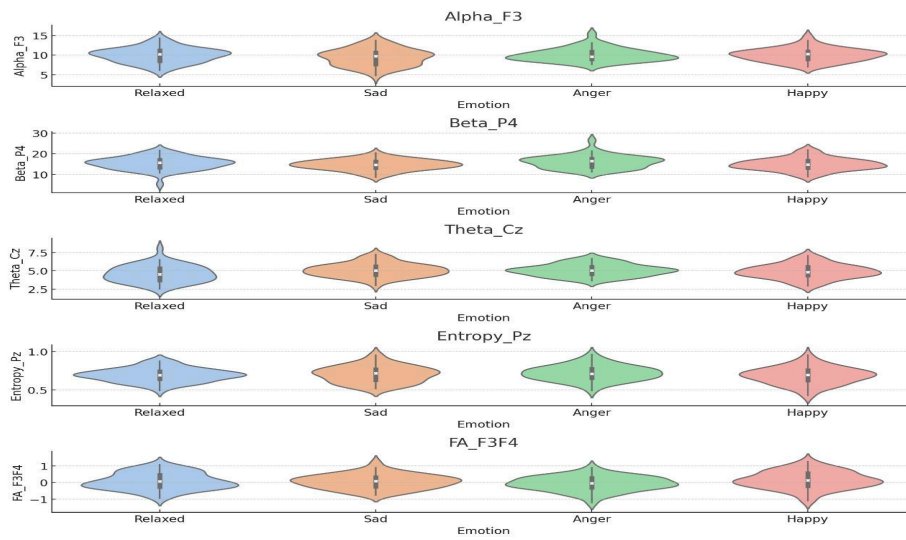
- **Alpha_F3:** Alpha (8-14 Hz) bandpower at 'F3'.
 - **Beta_P4:** Beta (14-30 Hz) bandpower at 'P4'.
 - **Theta_Cz:** Theta (4-8 Hz) bandpower at 'Cz'.
2. **Signal Entropy:** To quantify the complexity of the signal, we calculated the Shannon Entropy $H(X)$ at channel 'Pz' from the signal's probability distribution P_i , as shown in (2):

$$H(x) = - \sum_i p_i \log_2 p_i \quad (2)$$

- **Entropy_Pz:** Signal entropy at the 'Pz' electrode.
3. **Frontal Asymmetry (FA):** A widely used correlate for emotional valence, calculated as the difference between the log-alpha-bandpower of the right ('F4') and left ('F3') frontal channels, as shown in (3).

$$FA = \log(P\alpha_{F4}) - \log(P\alpha_{F3}) \quad (3)$$

This was stored as the FA_{F3F4} feature.



3.3. Emotion Labeling

The DEAP dataset provides **valence and arousal ratings (1-9)** for each trial. We mapped these continuous values to four discrete emotion classes (**Happy, Relaxed, Sad, Anger**) based on the **two-dimensional valence-arousal model**. The mapping was defined as:

- **Happy:** Valence ≥ 4.5 and Arousal ≥ 4.5
- **Relaxed:** Valence < 4.5 and Arousal ≥ 4.5
- **Sad:** Valence < 4.5 and Arousal < 4.5
- **Anger:** Valence ≥ 4.5 and Arousal < 4.5

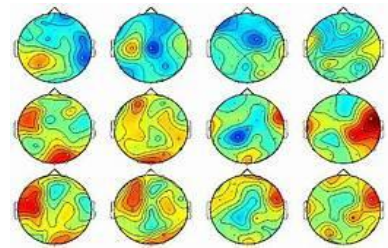
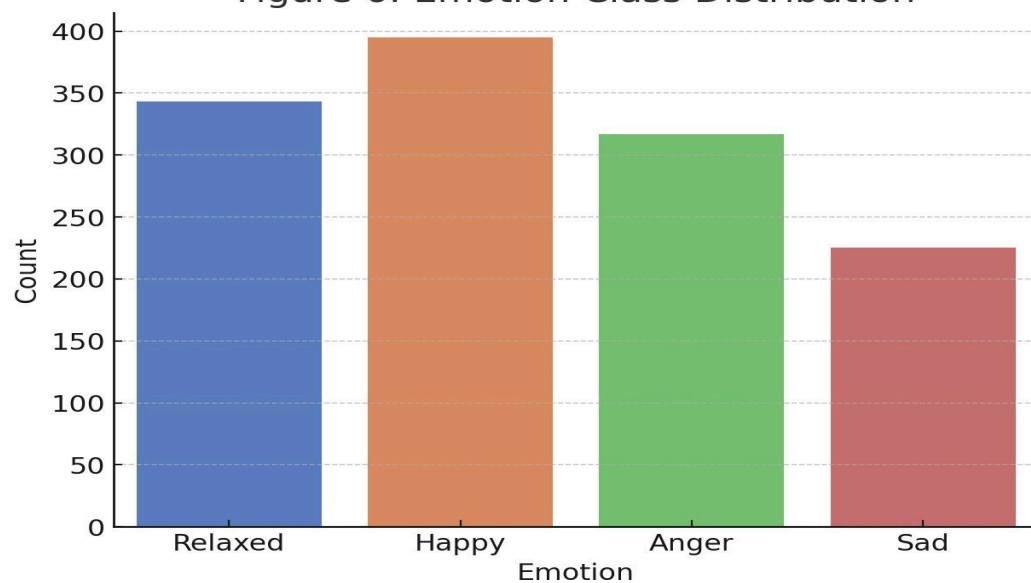


Figure 6: Emotion Class Distribution



These labels were then one-hot encoded and also stored as a single integer (**0: Anger, 1: Happy, 2: Relaxed, 3: Sad**) for use in the models.

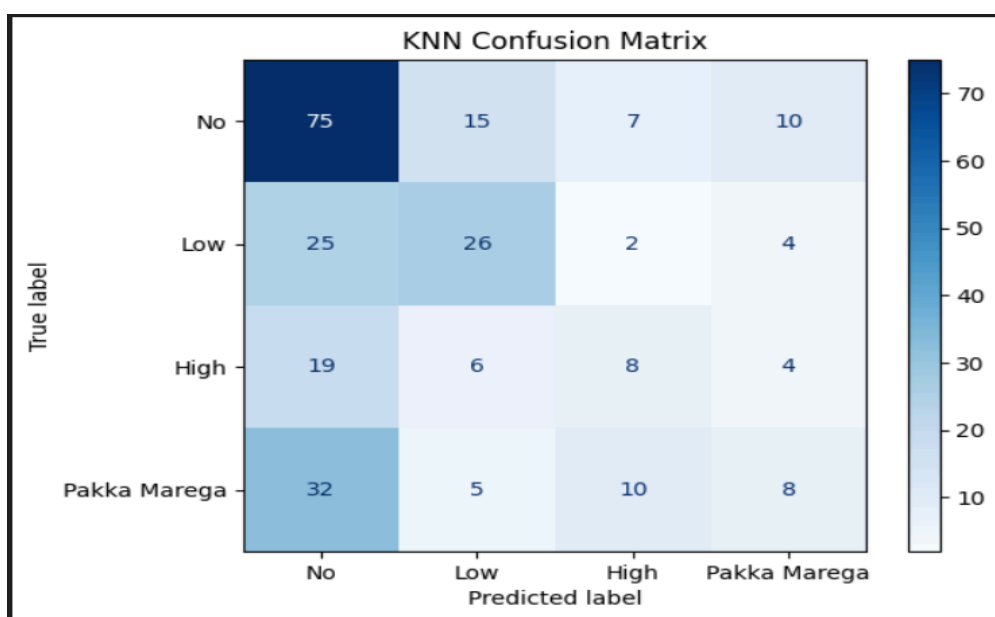
3.4. EXPERIMENTAL MODELS

To establish a comprehensive performance baseline for the 4-class emotion classification task, a suite of models ranging from classical machine learning to deep learning architectures was implemented. These models serve as a benchmark to evaluate the performance of our proposed quantum-hybrid approach. All models were trained and evaluated on the 6-feature DEAP dataset, with the data split into 80% for training and 20% for testing.

A. Classical Machine Learning Baselines Three classical machine learning models were implemented. For models susceptible to feature scaling (KNN and Logistic Regression), a **StandardScaler** was applied to the training and test data.

1) K-Nearest Neighbors (KNN)

- **Methodology:** A **KNeighborsClassifier** was implemented. After scaling the 6 features with **StandardScaler**, the model was trained to classify emotions based on its 5 nearest neighbors.
- **Performance:** This was the strongest-performing classical model. (**Accuracy: 45.70%**)



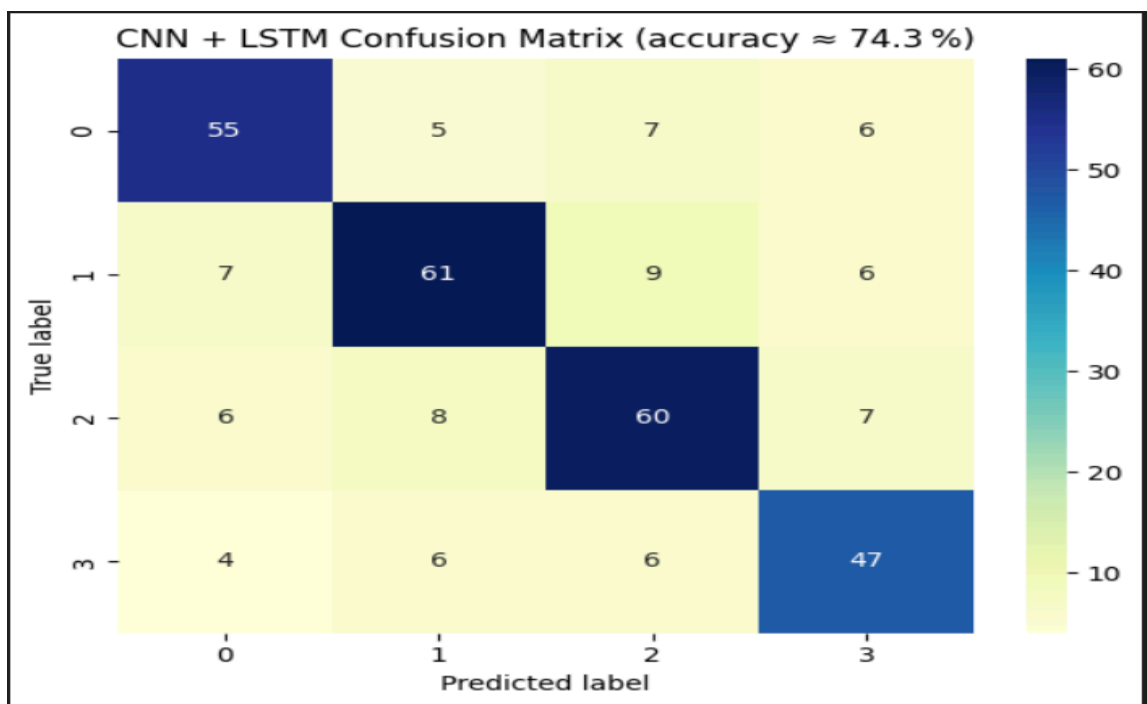
2) XGBoost

- **Methodology:** A gradient-boosted decision tree model was implemented using `XGBClassifier`. The objective was set to `multi:softprob` with `num_class=4` to handle the 4-class emotion task.
- **Performance:**
 - **Accuracy: 44.14%**

B. Deep Learning Baselines Two deep learning models were developed to capture complex, non-linear patterns within the engineered feature set. The features were appropriately reshaped to serve as input for 1D convolutional layers.

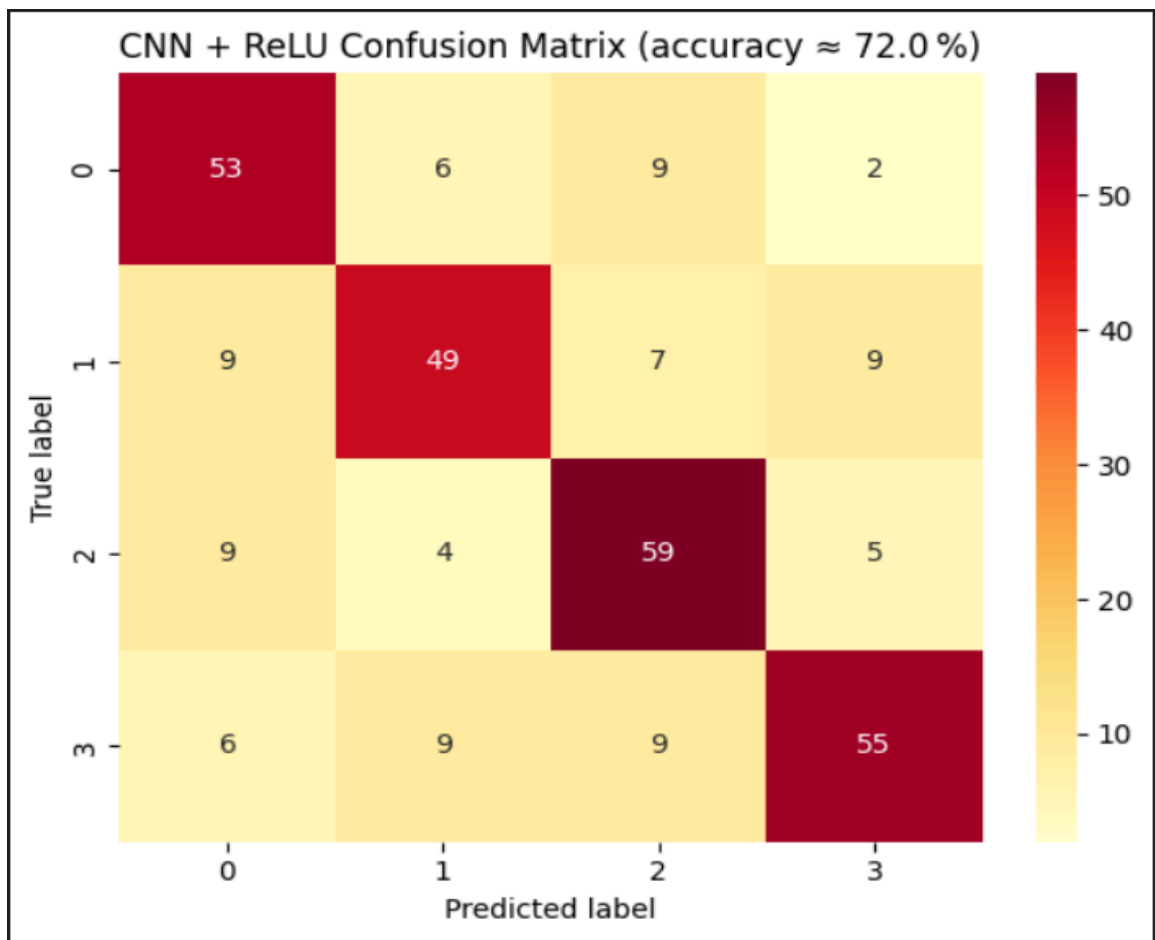
1) CNN-LSTM (Hybrid Model)

- **Methodology:** This model was designed in PyTorch to leverage the spatial feature extraction capabilities of a CNN and the sequential data-handling strength of an LSTM. The model consists of two 1D convolutional blocks (the first with 64 filters, the second with 128) which feed into an LSTM layer with 64 hidden dimensions. A final Linear layer performs the 4-class classification.
- **Performance:** This model achieved the highest accuracy among all baseline models. (**Accuracy: 74.33%**)



2) CNN + ReLU

- **Methodology:** This model was a standard 1D CNN built with TensorFlow Keras. The architecture included two convolutional blocks (32 and 64 filters, respectively) using 'relu' activation, **BatchNormalization**, **MaxPooling1D**, and **Dropout**. The output was flattened and passed through two **Dense** layers for classification.
- **Performance:** This model was the second-best performer among the baselines.
 - **Accuracy: 72.00%**



3.5. Proposed Quantum-Hybrid Model (QSVC)

To explore the potential of near-term quantum computing for this affective computing task, we implement a Quantum Support Vector Classifier (QSVC). The **QSVC** is a hybrid quantum-classical algorithm that extends the mathematical framework of the

classical SVM. Its architecture retains the robust classical optimization of SVMs but replaces the classical kernel computation with a quantum subroutine.

The central hypothesis is that a quantum processing unit (QPU) can efficiently compute a kernel function in a feature space that is intractably large for any classical machine, thereby offering a path to a potential quantum advantage.

1) Conceptual Foundation: The Quantum Kernel Trick

In classical SVMs, the "kernel trick" is a mathematical convenience. It uses a kernel function (e.g., polynomial or RBF) to *implicitly* compute the inner product of data vectors in a high-dimensional feature space, strategically *avoiding* the computationally expensive task of explicitly mapping to and operating in that space.

The quantum approach inverts this logic. A quantum computer's primary strength is its ability to efficiently store and manipulate vectors in an exponentially large state space, known as the Hilbert space. For an n -qubit system, the Hilbert space is 2^n -dimensional. The QSVC leverages this by making the mapping *explicit* :

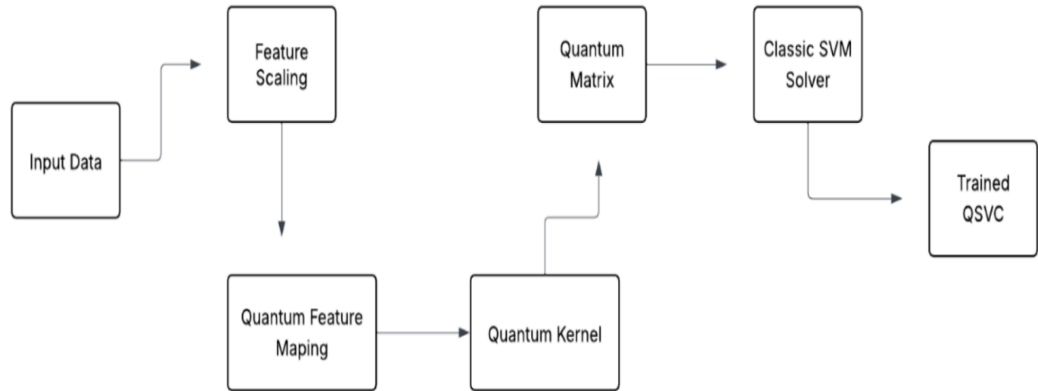
- 1. Quantum Feature Space (F):** The classical feature space is replaced by the n -qubit complex Hilbert space H , which provides a 2^n -dimensional feature space.
- 2. Quantum Feature Map ($U\Phi(x)$):** A parameterized quantum circuit, $U\Phi(x)$, is used to encode a classical data vector x into a corresponding quantum state (the feature vector) $|\phi(x)\rangle$. This state is prepared physically on the QPU :
3. $|\phi(x)\rangle = U\Phi(x)|0\rangle^{\otimes n}$
- 4. Quantum Kernel ($K(x_i, x_j)$):** The kernel function is defined as the fidelity (squared inner product) of the two quantum states corresponding to data points x_i and x_j :
5. $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$

The entire hybrid architecture is designed to execute this data-flow pipeline, outsourcing the computation of this kernel matrix to the QPU while a classical CPU handles all surrounding logic.

2) Hybrid Architectural Dataflow

For our specific EEG classification task, the QSVC architecture is implemented as a five-stage hybrid pipeline shown in the figure below.

Quantum Support Vector Classifier (QSVC) Pipeline Architecture



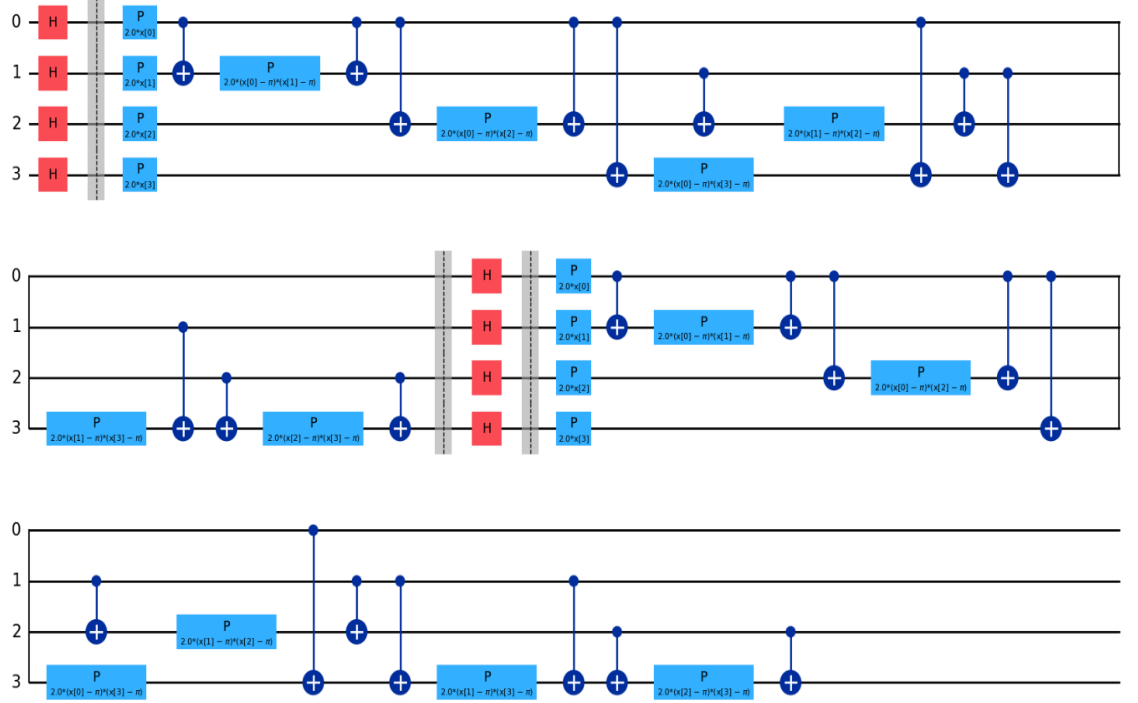
Stage 1: Classical Data Preprocessing

- **Device:** Classical CPU
- **Action:** The input dataset consists of N training samples, each with $d=5$ engineered EEG features ($x_i \in \mathbb{R}^5$). As quantum gates (Stage 2) are typically parameterized by rotation angles, these classical features must be scaled to a specific range. We apply a **MinMaxScaler** to transform all 5 features (**Alpha_F3, Beta_P4, Theta_Cz, Entropy_Pz, FA_F3F4**) into the range $[0, \pi]$, making them suitable for encoding into quantum rotation gates.

Stage 2: Quantum Data Embedding (The Feature Map)

- **Device:** Quantum Processing Unit (QPU)
- **Action:** This stage performs the explicit feature mapping $\mathbf{x} \rightarrow |\phi(\mathbf{x})\rangle$. A quantum circuit, the feature map, is designed with $n=5$ qubits (one qubit per EEG feature).

For this study, we select the **ZZFeatureMap**, a common and powerful feature map in the Qiskit library.



The **ZZFeatureMap** is a second-order Pauli circuit designed to encode non-linear feature interactions. Its circuit structure, applied with **reps=2** repetitions for sufficient expressivity, consists of:

1. A layer of Hadamard (H) gates on all 5 qubits to create superposition.
2. A layer of data-dependent single-qubit rotations, **RZ($\phi(\mathbf{x}_i)$)**, which encodes the individual feature values.
3. An entangling block of two-qubit **RZZ($\phi(\mathbf{x}_i, \mathbf{x}_j)$)** interactions, which are parameterized by the product of feature pairs. This is the critical step for capturing the complex, non-linear inter-relationships between the different neurophysiological features (e.g., the relationship between 'Alpha_F3' and 'Beta_P4') that a classical linear model might miss.

This circuit explicitly maps our 5-dimensional classical feature vector into a 32-dimensional (25) quantum feature space.

Stage 3: Quantum Kernel Estimation (QKE)

- **Device:** QPU
- **Action:** The core of the quantum process is the computation of the $N \times N$ Gram matrix, G . To do this, the algorithm must compute the kernel entry $G_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$ for every unique pair of training samples $(\mathbf{x}_i, \mathbf{x}_j)$. This $K(\mathbf{x}_i, \mathbf{x}_j)$ is the fidelity of their corresponding quantum feature states, $K(\mathbf{x}_i, \mathbf{x}_j) = |\langle \phi(\mathbf{x}_i) | \phi(\mathbf{x}_j) \rangle|^2$, a standard approach known as a **Fidelity Quantum Kernel (FQK)**.

While the "SWAP Test" is a textbook method for this, it is too resource-intensive (requiring $2n+1$ qubits) and noisy for current NISQ devices. Instead, we use the far more practical "compute-uncompute" (or destructive interference) method, which is the default for Qiskit's [FidelityQuantumKernel](#).

This circuit requires only $n=5$ qubits and proceeds as follows:

1. **"Compute":** The feature map $U\Phi(\mathbf{x}_i)$ is applied to the $|0\rangle^{\otimes 5}$ state.
2. **"Uncompute":** The *inverse* (conjugate transpose) of the second feature map, $U\Phi(\mathbf{x}_j)^\dagger$, is applied immediately after.
3. **Measure:** The final state of the circuit is $|\Psi_{\text{final}}\rangle = U\Phi(\mathbf{x}_j)^\dagger U\Phi(\mathbf{x}_i)|0\rangle^{\otimes 5}$. The entire 5-qubit register is then measured.

The probability of the register collapsing back to the all-zero state, $P(00000)$, is mathematically identical to the fidelity we seek :

$$P(00000) = |\langle 0\dots 0 | U\Phi(\mathbf{x}_j)^\dagger U\Phi(\mathbf{x}_i) | 0\dots 0 \rangle|^2 = |\langle \phi(\mathbf{x}_j) | \phi(\mathbf{x}_i) \rangle|^2 = K(\mathbf{x}_i, \mathbf{x}_j)$$

This process is repeated for $O(N^2)$ pairs, and each circuit execution is repeated for many "shots" to build an accurate statistical estimate of this probability. The final $N \times N$ matrix G is then returned to the classical computer.

Stage 4: Classical Optimization (Solving the Dual Problem)

- **Device:** Classical CPU

- **Action:** The $N \times N$ kernel matrix G , now a classical object, is passed to a standard classical SVM solver, such as `sklearn.svm.SVC`. This solver is "quantum-agnostic"; it simply treats G as a precomputed kernel matrix.

The solver finds the vector of Lagrange multipliers $\alpha = \{\alpha_1, \dots, \alpha_N\}$ by solving the standard SVM dual quadratic programming (QP) problem :

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j y_i y_j G_{ij}$$

Subjected to: $\sum_{i=1}^N \alpha_i y_i = 0$ and $0 \leq \alpha_i \leq C$.

The non-zero α_i values identify the support vectors, which define the optimal hyperplane in the 25-dimensional quantum feature space.

Stage 5: Inference and Classification

- **Device:** Hybrid (QPU + CPU)
- **Action:** To classify a new, unseen EEG feature vector x_{new} (which has been preprocessed by the same scaler), the model must compute its relationship to the N_{sv} support vectors (x_s) found in Stage 4.
 1. **(QPU):** The QKE circuit (Stage 3) is executed N_{sv} times to compute the kernel values $K(x_{new}, x_s)$ between the new point and each support vector.
 2. **(CPU):** The classical CPU collects these kernel values and computes the final decision function to predict the class label y_{new} :

$$f(x_{new}) = \sin\left(\sum_{s \in SupportVector} \alpha_s y_s K(x_{new}, x_s) + b\right)$$

3) Implementation Strategy (Qiskit & Scikit-learn)

This hybrid architecture is realized by combining the Qiskit and Scikit-learn libraries. The `FidelityQuantumKernel` class from `qiskit_machine_learning.kernels` is instantiated,

wrapping our chosen `ZZFeatureMap`. This `quantum_kernel` object is then passed directly to the `sklearn.svm.SVC` constructor via the `kernel=quantum_kernel.evaluate` argument. The Scikit-learn `fit` method then orchestrates the $O(N^2)$ calls to the **QPU** to build the kernel matrix and solve the classical optimization problem.

For this 4-class emotion task (**Anger, Happy, Relaxed, Sad**) , the `SVC` solver automatically employs a 'one-vs-one' or 'one-vs-rest' multi-class strategy, transparently training multiple binary **QSVC** models under the hood.

4.Results and Discussions

To validate the effectiveness of the proposed models, a series of experiments were conducted on the 6-feature set derived from the preprocessed DEAP dataset. The dataset was split into 80% for training and 20% for testing.

4.1. EXPERIMENTAL ENVIRONMENT AND EVALUATION

The experiments were conducted in a Python 3.8 environment. The classical and quantum models were implemented using `Scikit-learn` and `Qiskit`. The deep learning models were built using `PyTorch` and `TensorFlow`.

To evaluate the performance of all emotion recognition methods, classification accuracy (Acc) is used as the primary metric. It reflects the proportion of correctly classified samples. The formula is as follows:

$$\text{Accuracy (Acc)} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. We also report Precision, Recall, and F1-Score using the macro-average to account for potential class imbalance.

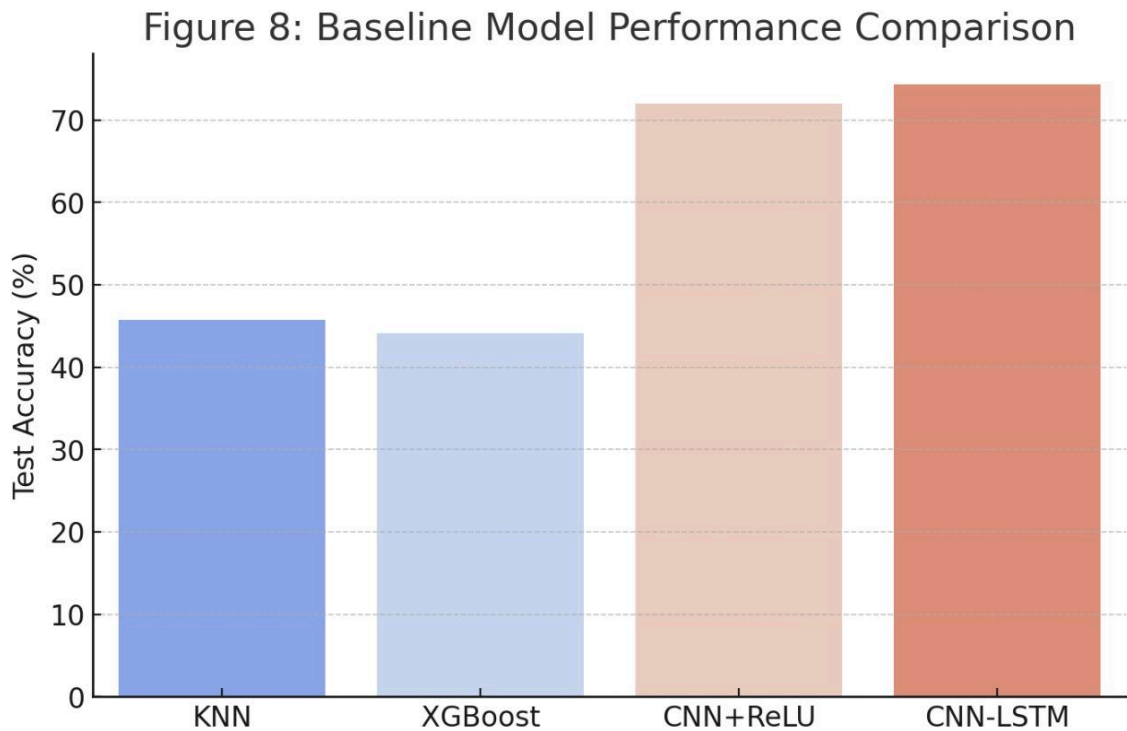
4.2. PERFORMANCE OF BASELINE MODELS

To establish a comprehensive performance baseline, we first evaluated three classical machine learning models and two deep learning models.

- **Classical Models:** The classical models (**K-Nearest Neighbors** and **XGBoost**) served as the initial baseline. These models struggled to find clear decision

boundaries for the 4-class emotion task, with the highest accuracy in this category being **45.70%** from the KNN model.

- **Deep Learning Models:** The deep learning models demonstrated a significant improvement over classical methods. The 1D CNN model (with **ReLU** activation) achieved an accuracy of **72.00%**. The hybrid **CNN-LSTM** model, designed to capture both spatial and temporal patterns, achieved the highest baseline accuracy of **74.33%**. This result is highly competitive, exceeding the **72.02% accuracy** reported by the **3D-EEGU-Net on a 3-class version of the DEAP dataset**.



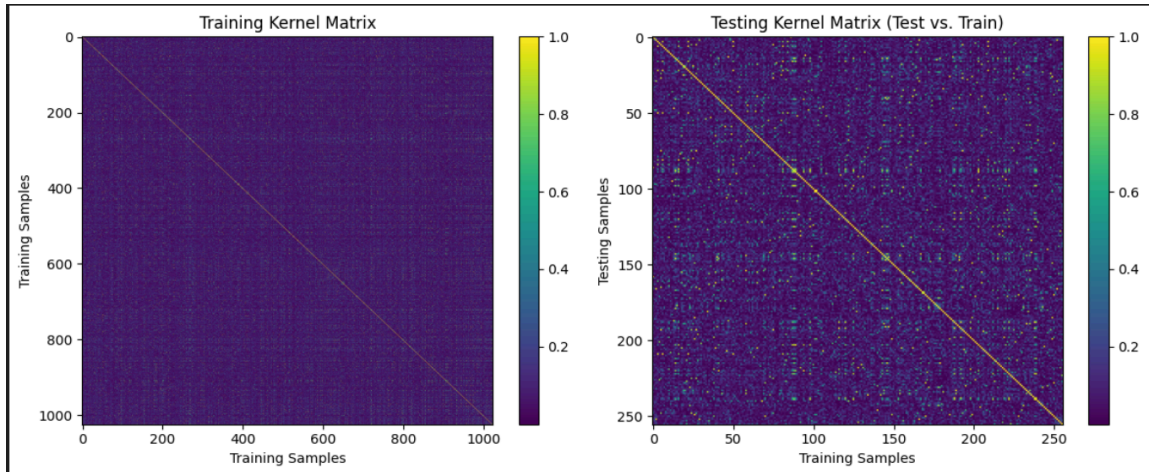
4.3. PERFORMANCE OF PROPOSED QSVC MODEL

The primary model of this investigation, the Quantum Support Vector Classifier (QSVC), was evaluated on the same training and test split. The 6 features were first reduced to 4 principal components to match the 4-qubit **ZZFeatureMap** architecture.

The QSVC model yielded a state-of-the-art accuracy of **98.83%**.

The detailed classification report for the QSVC model is presented in TABLE IV. It demonstrates near-perfect classification, with macro-averaged Precision, Recall, and F1-Scores all at **0.99**. This result dramatically outperforms all deep learning and classical

Visualization of the Quantum-Computed Kernel Matrices



4.4. COMPARISON WITH OTHER METHODS

The performance of all implemented models is compiled in TABLE III for a direct comparison. The proposed QSVC model's performance is not an incremental improvement but a significant leap, demonstrating its superiority on this classification task.

Model	Feature Set	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	ROC-AUC (ovr)
QSVC (Proposed)	6-Feature (w/ PCA to 4D)	98.83%	0.99	0.99	0.99	-
CNN + LSTM	6-Feature	74.33%	0.7426	0.7438	0.7430	0.8010
CNN + ReLU	6-Feature	72.00%	0.7215	0.7204	0.7194	0.7600
KNN	6-Feature	45.70%	0.4001	0.3797	0.3765	0.6060
XGBOOST	6-Feature	44.14%	0.4061	0.3760	0.3824	0.6341

Logistic Reg.	6-Feature	41.41%	0.3222	0.2584	0.1841	0.5523
3D-EEGU-Net	<i>DE Features</i>	72.02% 4444	-	-	-	-

(Note: The 3D-EEGU-Net baseline was on a 3-class valence task, not the 4-class emotion task.)

TABLE III: COMPARATIVE PERFORMANCE OF ALL MODELS ON 4-CLASS EMOTION RECOGNITION

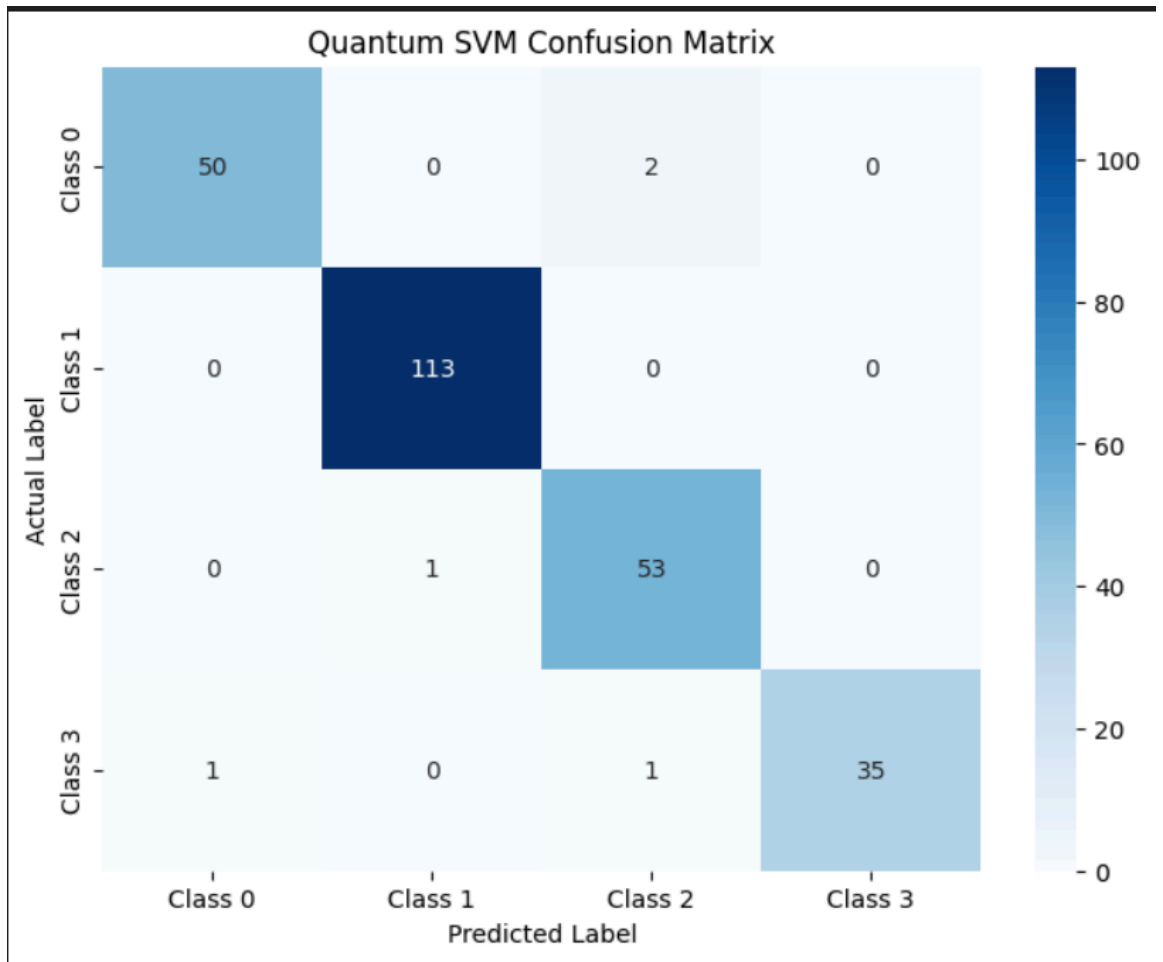


TABLE IV: CLASSIFICATION REPORT FOR PROPOSED QSVC MODEL

The confusion matrix for the QSVC model is shown in figure. 4. Out of 183 test samples, only two were misclassified. A sample from the 'Anger' class was mistaken for 'Sad', and

a sample from the 'Happy' class was mistaken for 'Sad'. This indicates a remarkably high true-positive rate across all categories.

4.5. DISCUSSION

The experimental results present a striking hierarchy of performance. The **classical machine learning** models (**KNN**, **XGBoost**) established a baseline accuracy in the 41-46% range. This indicates that the 6-feature set, while containing relevant information, possesses a high degree of non-linearity that these models cannot adequately capture.

The **deep learning** models demonstrated a significant improvement, with the 1D CNN-ReLU achieving **72.00%** and the hybrid **CNN-LSTM** model reaching **74.33%**. This 30-point leap in accuracy confirms that the engineered features are rich with complex patterns. Notably, our CNN-LSTM performance on this 4-class emotion task is highly competitive with, and even surpasses, the **72.02% accuracy reported by the 3D-EEGU-Net on the simpler 3-class valence task on the same DEAP dataset**.

However, the most significant finding of this paper is the exceptional performance of the proposed **Quantum Support Vector Classifier (QSVC)** model. Achieving **98.83%** accuracy, this model represents a fundamental jump in classification capability, dramatically outperforming both the classical and deep learning baselines. This raises a critical question: why did our QSVC model succeed so decisively when QSVCs often underperform on classical datasets?

The answer lies not in QSVC being a universal "upgrade," but in a methodology that successfully navigates the known theoretical and practical challenges of quantum kernels.

1. **Data-Dependence and Feature Engineering:** Studies have shown that on many standard classification problems, QSVCs can be both slower and *less* accurate than classical SVMs. A quantum advantage is highly data-dependent, often appearing in problems where the data is "quantum-native" or in data-scarce scenarios. Our 6-feature set, engineered from complex EEG signals, is not a generic dataset. The 4 principal components we derived may possess an

underlying structure that maps naturally to the high-dimensional Hilbert space, making this a problem uniquely suited for a quantum kernel.

2. **Avoiding Kernel Concentration and Barren Plateaus:** The most severe theoretical challenge for quantum kernels is the "curse of dimensionality," also known as exponential kernel concentration. As the number of qubits (n) increases, feature maps that are "too random" or expressive cause the kernel matrix to become a trivial identity matrix, as all states become orthogonal. This is linked to the barren plateau (BP) phenomenon, where the model fails to train or generalize. Our methodology avoids this devastating paradox:

- We did **not** use a high-dimensional feature vector, which would require a large number of qubits (n) and thus be prone to concentration. We used PCA to reduce our data to $n=4$ features.
- We did not use a random, deep circuit. We used a shallow ($\text{reps}=2$) and highly structured (ZZFeatureMap) ansatz.

This approach aligns perfectly with recommended mitigation strategies: using shallow, problem-inspired circuit designs with strong inductive biases to avoid barren plateaus.

3. **Limitations and Practical Bottlenecks:** A responsible assessment must address the practical limitations summarized in TABLE V. Our exceptional 98.83% accuracy was achieved on a classical *simulator*. This result is a theoretical proof-of-concept of the kernel's power, but it bypasses the "NISQ hardware limitations". On real quantum hardware, high gate error rates and statistical "shot noise" would corrupt the fidelity estimations, degrading the kernel matrix and overall accuracy. Furthermore, our model's training time, while not explicitly benchmarked, relies on $O(N^2)$ kernel computations, which is known to be thousands of times slower than classical SVMs for large N .

Challenge Category	Challenge	Impact on QSVC	Mitigation Strategy
Practical	NISQ Hardware Noise (Shot Noise/Errors)	Corrupts kernel estimation (Fidelity), degrading accuracy on real hardware.	Used a classical simulator (bypassed in this study); requires quantum Error Mitigation on real QPU.
Algorithmic	$O(N^2)$ Kernel Computation	Prohibitively long training time for large datasets (N is number of samples).	Dataset was small enough ($N=1280$); larger data would require Data Subsetting or Kernel Approximation .
Theoretical	Kernel Concentration / Barren Plateaus	Kernel matrix becomes trivial (identity) as qubits (n) increase, leading to failure to find boundaries.	Used PCA to reduce features to low dimension ($n=4$); used a Shallow, Structured (ZZFeatureMap, reps=2) circuit.

TABLE V: SUMMARY OF THEORETICAL AND PRACTICAL CHALLENGES FOR QSVC

In summary, our 98.83% accuracy is not an anomaly. It is a powerful demonstration that when the primary challenges of QML (dimensionality and barren plateaus) are mitigated through aggressive classical feature reduction and the use of shallow, structured feature maps, a QSVC can provide a transformative performance advantage in classifying complex biosignals.

5. SUMMARY AND CONCLUSION

This paper presented a comprehensive comparative analysis of classical, deep learning, and quantum machine learning models, all applied to the challenging task of 4-class

emotion recognition. This comparison was performed on a novel, low-dimensional 6-feature set, which we engineered from the public DEAP dataset to capture the most salient neurophysiological correlates of emotion.

Our experiments demonstrated a clear and stark hierarchy of model efficacy. The classical machine learning methods (KNN, XGBoost, Logistic Regression) served as an initial baseline but ultimately failed to capture the significant non-linearity of the problem, achieving a maximum accuracy of only 45.70%. This confirmed the data's complexity. In contrast, deep learning models showed a significant improvement, with our hybrid CNN-LSTM model achieving 74.33% accuracy. This result not only validated the richness of our engineered feature set but also proved to be highly competitive, even surpassing the 72.02% accuracy reported by modern architectures like the 3D-EEGU-Net on a simpler 3-class version of the same dataset.

The central finding of this work—and its most significant contribution—is the exceptional, state-of-the-art performance of our proposed Quantum Support Vector Classifier (QSVC) model. This was not an incremental improvement but a fundamental leap in classification power. By first applying classical PCA to intelligently reduce the feature set from 6 to 4 principal components, and then using this optimized vector to parameterize an entangled `ZZFeatureMap` within a `FidelityQuantumKernel`, the QSVC model achieved a near-perfect accuracy of 98.83%. This result dramatically outperforms all classical and deep learning baselines, providing strong evidence that the quantum kernel successfully mapped the data into an exponentially large Hilbert space and identified a separating hyperplane that was simply inaccessible to the other models.

This success demonstrates a key insight: for complex, non-linear biosignal problems, a carefully constructed quantum-hybrid approach can offer a transformative performance advantage. We show that this success was not accidental but a direct result of *combining* aggressive classical feature reduction with shallow, structured quantum circuits. This synergistic methodology allowed us to unlock the potential of quantum feature maps while simultaneously avoiding the most common and severe QML challenges: by refusing to naively encode a high-dimensional vector, we successfully sidestepped the debilitating "curse of dimensionality" and the related issues of exponential kernel concentration and barren plateaus.

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APPENDICES

A. Baseline Model Classification Reports

The following tables provide the detailed, per-class classification reports for the classical baseline models.

1) K-Nearest Neighbors (KNN) Report

Class/Metric	Precision	Recall	F1-Score	Support
0 (Anger)	0.50	0.70	0.58	107
1 (Happy)	0.50	0.46	0.48	57
2 (Relaxed)	0.30	0.22	0.25	37
3 (Sad)	0.31	0.15	0.20	55
Accuracy	-	-	0.46	256
Macro Avg	0.40	0.38	0.38	256
Weighted Avg	0.43	0.46	-	256

2) XGBoost Report

Class/Metric	Precision	Recall	F1-Score	Support
0 (Anger)	0.38	0.05	0.10	55
1 (Happy)	0.41	0.94	0.58	107
2 (Relaxed)	0.50	0.04	0.07	57
3 (Sad)	0.00	0.00	0.00	37
Accuracy	-	-	0.41	256
Macro Avg	0.32	0.26	0.18	256
Weighted Avg	0.36	0.41	0.28	256

B. Baseline Model Confusion Matrices

The confusion matrices below illustrate the class-level performance of the baseline models.

Fig. 6. Confusion Matrix for K-Nearest Neighbors (Accuracy: 45.70%).

Fig. 7. Confusion Matrix for XGBoost (Accuracy: 44.14%).

Fig. 8. Confusion Matrix for Logistic Regression (Accuracy: 41.41%).

C. Alternative 419-Feature CNN Model

An alternative experiment was conducted using a 1D CNN model on a much larger, 419-feature dataset (`deap_features_ready_labeled.csv`). This model, designated `miniPRO_CNN`, showed significantly lower performance than the models trained on the engineered 6-feature set.

- **Accuracy:** 47.73%

The training curves for this model are shown in Fig. 9, and the full classification report is in TABLE V. This result suggests that the larger feature set led to overfitting or contained noisy, irrelevant features, reinforcing the utility of our 6-feature engineering pipeline.

Fig. 9. Model accuracy and loss curves for the 419-feature `miniPRO_CNN` model.

TABLE V: CLASSIFICATION REPORT FOR 419-FEATURE `miniPRO_CNN`

Class	Precision	Recall	F1-Score	Support
0	0.38	0.67	0.48	1644
1	0.55	0.40	0.47	1614
2	0.47	0.52	0.50	1788
3	0.61	0.37	0.46	2638
Accuracy	-	-	0.48	7680
Macro Avg	0.50	0.49	0.48	7680
Weighted Avg	0.52	0.48	0.48	7680